Expt No: 1
Date : 16/07/26

BASIC NETWORKING COMMANDS.

Aim :-

Basic Network Commands in Linux and Windows Operating Systems.

Basic Networking Command :-

arp -a :- ARP is short form of address resolution protocol.

hostname :- This is the simplest of all TCP / IP Commands

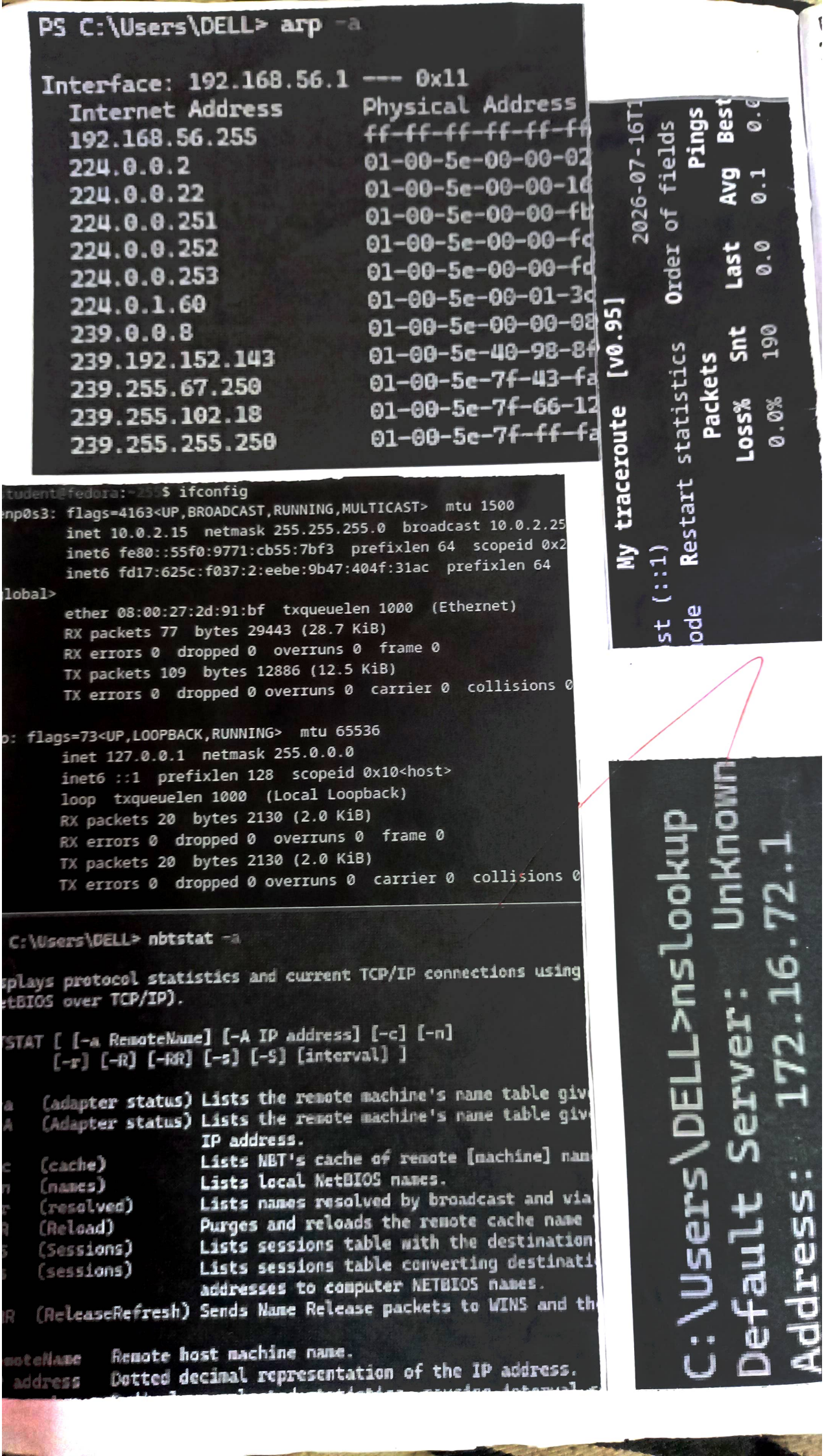
ipconfig / all :- This Command displays detailed configuration information about Connection including Router, Gateway, DNS, DHCP.

nbtstat -a :- This Command helps Solve problems with NetBIOS name resolution for NetBIOS over TCP/

netstat -a :- This is a tool used to perform DNS lookups in DNS details.

nslookup :- This Retrive the IP address of a do Name or DNS Server details.

pathping : This is to trace the route to a destin and checks packet loss at each hop.



```
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address
  192.168.56.255        ff-ff-ff-ff-ff-ff
  224.0.0.2             01-00-5e-00-00-02
  224.0.0.22            01-00-5e-00-00-16
  224.0.0.251           01-00-5e-00-00-fb
  224.0.0.252           01-00-5e-00-00-fc
  224.0.0.253           01-00-5e-00-00-fd
  224.0.1.60            01-00-5e-00-01-3c
  239.0.0.8             01-00-5e-00-00-08
  239.192.152.143       01-00-5e-40-98-8f
  239.255.67.250        01-00-5e-7f-43-fa
  239.255.102.18        01-00-5e-7f-66-12
  239.255.255.250       01-00-5e-7f-ff-fa
```

```
My traceroute  [v0.95]                    2026-07-16T
st (::1)                                  Order of fields
ode   Restart statistics                        Pings
                    Packets                Last  Avg  Best
                 Loss%   Snt               0.0   0.1  0.0
                 0.0%    190
```

```
tudent@fedora:~$ ifconfig
np0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
       inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.25
       inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x2
       inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64
lobal>
       ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
       RX packets 77  bytes 29443 (28.7 KiB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 109  bytes 12886 (12.5 KiB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

o: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
       inet 127.0.0.1  netmask 255.0.0.0
       inet6 ::1  prefixlen 128  scopeid 0x10<host>
       loop  txqueuelen 1000  (Local Loopback)
       RX packets 20  bytes 2130 (2.0 KiB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 20  bytes 2130 (2.0 KiB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
C:\Users\DELL> nbtstat -a

splays protocol statistics and current TCP/IP connections using
etBIOS over TCP/IP).

TSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
      [-r] [-R] [-RR] [-s] [-S] [interval] ]

a   (adapter status) Lists the remote machine's name table giv
A   (Adapter status) Lists the remote machine's name table giv
                     IP address.
c   (cache)          Lists NBT's cache of remote [machine] nam
n   (names)          Lists local NetBIOS names.
r   (resolved)       Lists names resolved by broadcast and via
R   (Reload)         Purges and reloads the remote cache name
S   (Sessions)       Lists sessions table with the destination
s   (sessions)       Lists sessions table converting destinati
                     addresses to computer NETBIOS names.
RR  (ReleaseRefresh) Sends Name Release packets to WINS and th

emoteName   Remote host machine name.
 address    Dotted decimal representation of the IP address.
```

```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

Ping: This Sends ICMP packet to check Whether a host is reachable and measures response time.

route: Displays or Changes the IP routing tables used by the System.

Linux Networking Commands:-

* ip: The ip Command is one of the basic Commands every administrator will need in daily work. The ip Command Can Show address information manipulate routing plus display network Various, integer and tunnels.

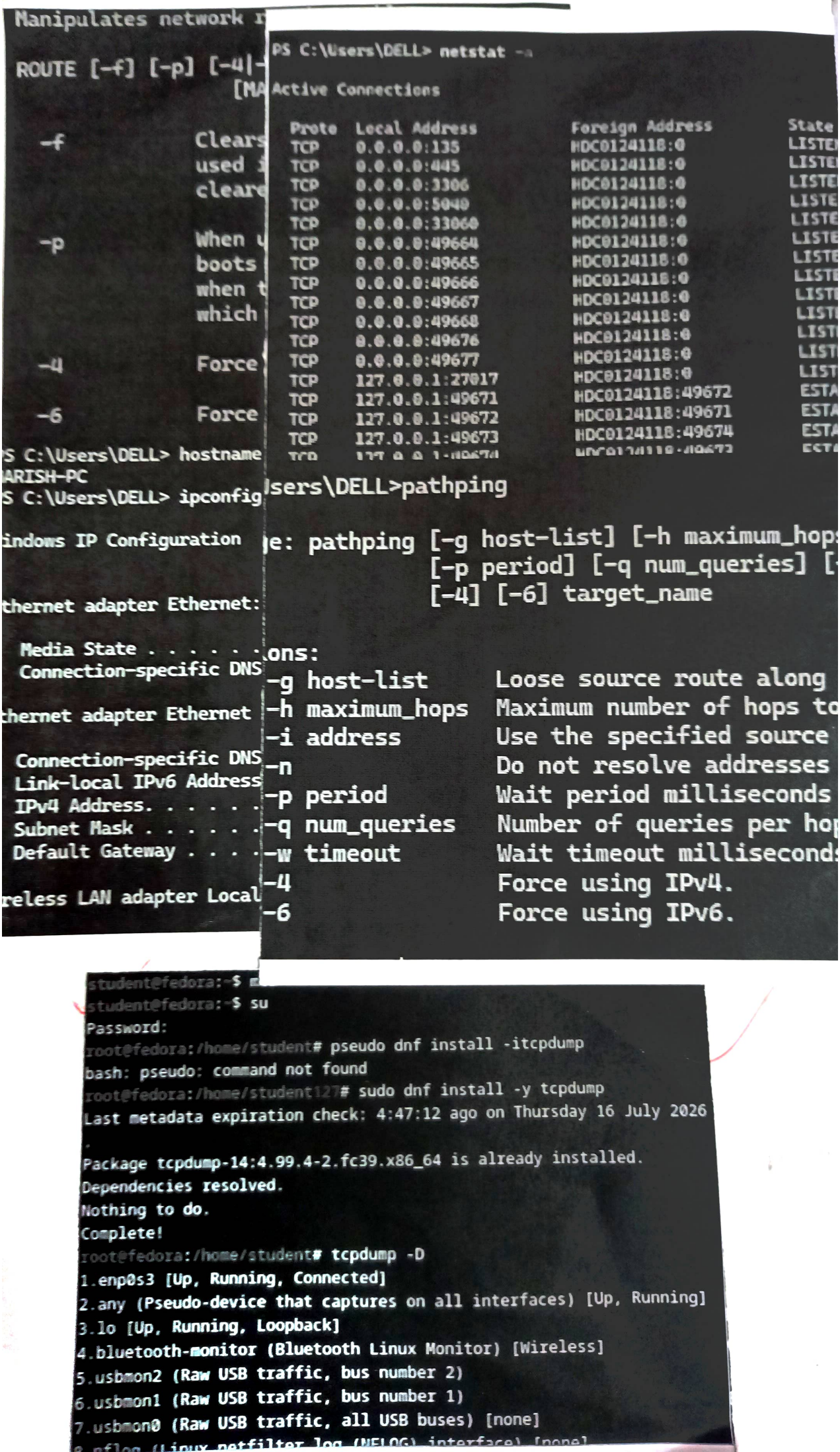
* ifconfig :- The if Config Command is a Staple in many Sysadmin's tool belt for Configuration and troubleshooting

* mtr :- It is a program with a Command-line interface The Serves as a network diagonistic and trouble Shooting tool.

* tcpdump:-

The tcpdump Command is designed for Capturing and displaying packets.

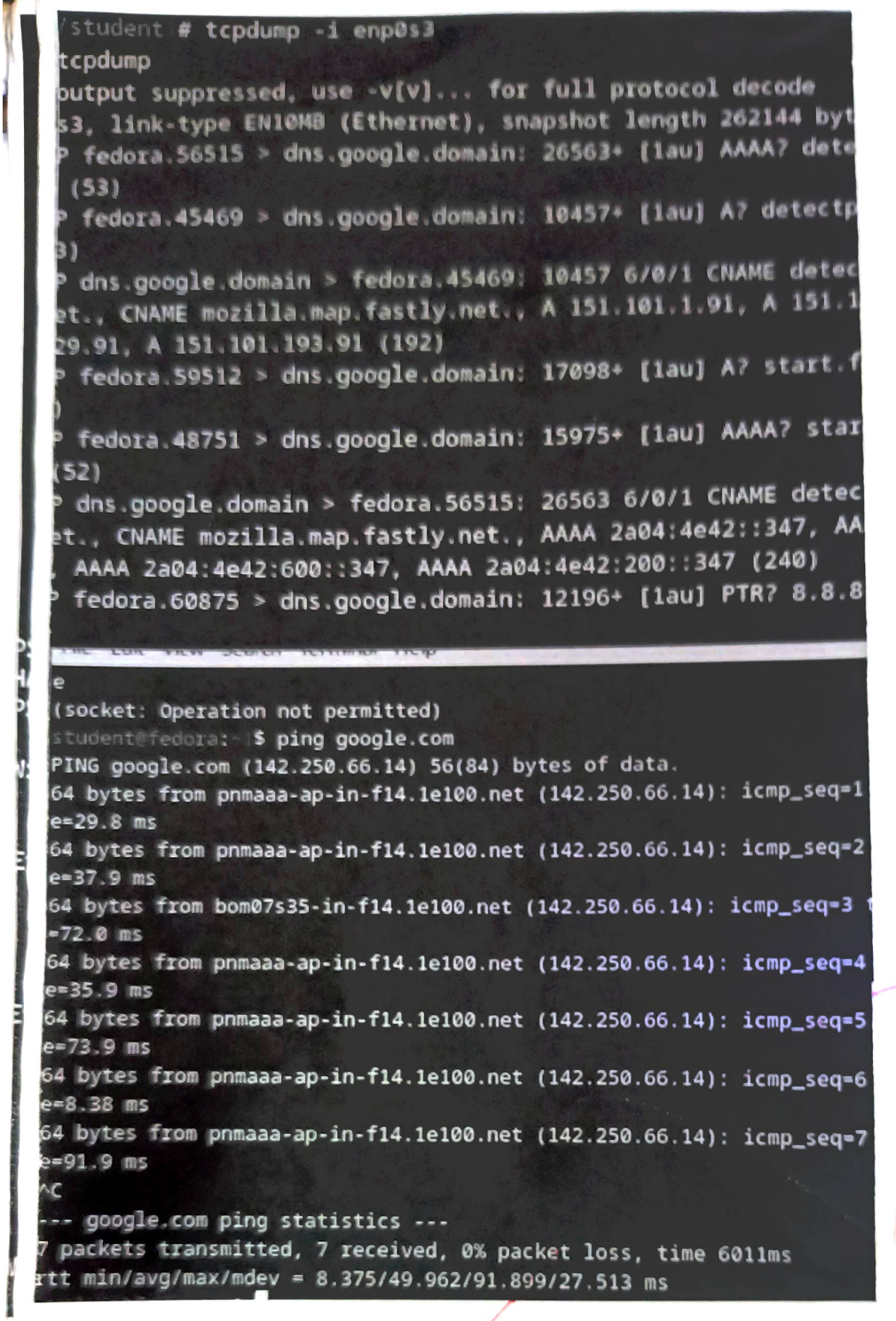
* Ping:- It is a tool that verifies IP-Level Connectivity another TCP/IP Computer by Sending Internet Control message Protocol (ICMP) Echo Request message.


Manipulates network
ROUTE [-f] [-p] [-4|-
[MA
-f
Clears
used
cleare
-p
When
boots
when
which
-4
Force
-6
Force
S C:\Users\DELL> hostname
ARISH-PC
S C:\Users\DELL> ipconfig
indows IP Configuration
thernet adapter Ethernet:
Media State . . . . . .
Connection-specific DNS
thernet adapter Ethernet
Connection-specific DNS
Link-local IPv6 Address
IPv4 Address. . . . . .
Subnet Mask . . . . . .
Default Gateway . . . .
reless LAN adapter Local
PS C:\Users\DELL> netstat -a
Active Connections
Prote Local Address Foreign Address State
TCP 0.0.0.0:135 HDC0124118:0 LISTE
TCP 0.0.0.0:445 HDC0124118:0 LISTE
TCP 0.0.0.0:3306 HDC0124118:0 LISTE
TCP 0.0.0.0:5040 HDC0124118:0 LISTE
TCP 0.0.0.0:33060 HDC0124118:0 LISTE
TCP 0.0.0.0:49664 HDC0124118:0 LISTE
TCP 0.0.0.0:49665 HDC0124118:0 LISTE
TCP 0.0.0.0:49666 HDC0124118:0 LISTE
TCP 0.0.0.0:49667 HDC0124118:0 LISTE
TCP 0.0.0.0:49668 HDC0124118:0 LISTE
TCP 0.0.0.0:49676 HDC0124118:0 LISTE
TCP 0.0.0.0:49677 HDC0124118:0 LISTE
TCP 127.0.0.1:27017 HDC0124118:0 LIST
TCP 127.0.0.1:49671 HDC0124118:49672 ESTA
TCP 127.0.0.1:49672 HDC0124118:49671 ESTA
TCP 127.0.0.1:49673 HDC0124118:49674 ESTA
sers\DELL>pathping
e: pathping [-g host-list] [-h maximum_hops
[-p period] [-q num_queries] [
[-4] [-6] target_name
ons:
-g host-list Loose source route along
-h maximum_hops Maximum number of hops to
-i address Use the specified source
-n Do not resolve addresses
-p period Wait period milliseconds
-q num_queries Number of queries per hop
-w timeout Wait timeout milliseconds
-4 Force using IPv4.
-6 Force using IPv6.
student@fedora:~$ su
Password:
root@fedora:/home/student# pseudo dnf install -itcpdump
bash: pseudo: command not found
root@fedora:/home/student127# sudo dnf install -y tcpdump
Last metadata expiration check: 4:47:12 ago on Thursday 16 July 2026
.
Package tcpdump-14:4.99.4-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@fedora:/home/student# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]

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Result:

Basic networking Commands in both Linux and Windows operating System is successfully executed and verified.



```
/student # tcpdump -i enp0s3
tcpdump
output suppressed, use -v[v]... for full protocol decode
s3, link-type EN10MB (Ethernet), snapshot length 262144 byt
P fedora.56515 > dns.google.domain: 26563+ [1au] AAAA? dete
 (53)
P fedora.45469 > dns.google.domain: 10457+ [1au] A? detectp
3)
P dns.google.domain > fedora.45469: 10457 6/0/1 CNAME detec
et., CNAME mozilla.map.fastly.net., A 151.101.1.91, A 151.1
29.91, A 151.101.193.91 (192)
P fedora.59512 > dns.google.domain: 17098+ [1au] A? start.f
)
P fedora.48751 > dns.google.domain: 15975+ [1au] AAAA? star
(52)
P dns.google.domain > fedora.56515: 26563 6/0/1 CNAME detec
et., CNAME mozilla.map.fastly.net., AAAA 2a04:4e42::347, AA
, AAAA 2a04:4e42:600::347, AAAA 2a04:4e42:200::347 (240)
P fedora.60875 > dns.google.domain: 12196+ [1au] PTR? 8.8.8
```

```
e
(socket: Operation not permitted)
student@fedora:~$ ping google.com
PING google.com (142.250.66.14) 56(84) bytes of data.
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1
e=29.8 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2
e=37.9 ms
64 bytes from bom07s35-in-f14.1e100.net (142.250.66.14): icmp_seq=3 t
=72.0 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4
e=35.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5
e=73.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6
e=8.38 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7
e=91.9 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 8.375/49.962/91.899/27.513 ms
```

```
udent@fedora:~$ ip
sage: ip [ OPTIONS ] OBJECT { COMMAND | help }
      ip [ -force ] -batch filename
here  OBJECT := { address | addrlabel | amt | fou | help | ila |

                 link | macsec | maddress | monitor | mptcp | mr

                 neighbor | neighbour | netconf | netns | nextho
                 ntbl | route | rule | sr | tap | tcpmetrics |
                 token | tunnel | tuntap | vrf | xfrm }
    OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[e
                 -h[uman-readable] | -iec | -j[son] | -p[retty]
                 -f[amily] { inet | inet6 | mpls | bridge | lin
                 -4 | -6 | -M | -B | -0 |
                 -l[oops] { maximum-addr-flush-attempts } | -br
                 -o[neline] | -t[imestamp] | -ts[hort] | -b[atc

                 -rc[vbuf] [size] | -n[etns] name | -N[umeric]
                 -c[olor]}
```